Class, Object, Methods

20 June 2026 11:26

Till now we have written most of the code inside :

```
class ConBreak
{
    public static void main(String[] args)
```

We have learned:

1. Variables
2. Operators
3. Conditions
4. Loops

But after doing so many programs, till now we learned how to organize data and how java behaves.

What is class ?

Class defines what an object should have and what it should do.

What is object ?

Object is the real instance created from class

What is method ?

Method represents an action or behaviour

What does a Student have ? What can a student do ?

Student has:

1. Name
2. Age
3. Roll num
4. Course

Student can:

1. Study
2. Attend class
3. Take exam
4. Notes

Let's connect this with java

Data → becomes fields	Action → becomes methods

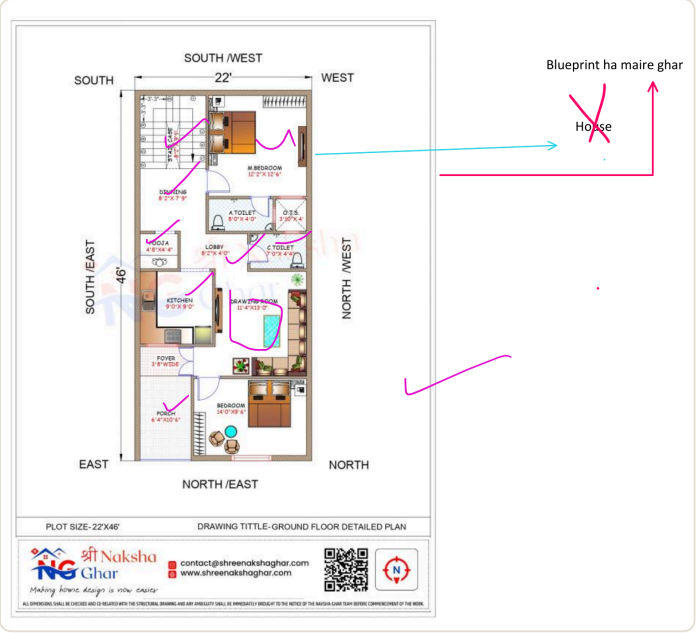
Both are placed inside class

01 What is class ?

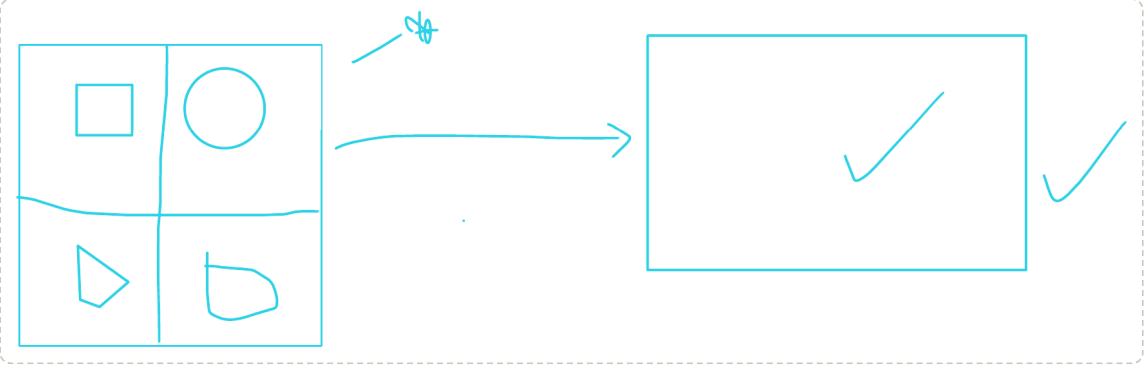
> A class is a **blueprint** or **template** used to create objects

SIMPLE DEFINITION:

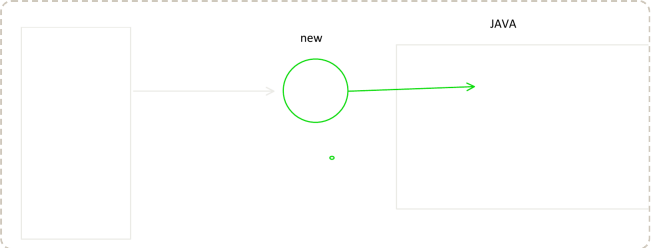
Class = blueprint that defines data and behaviours

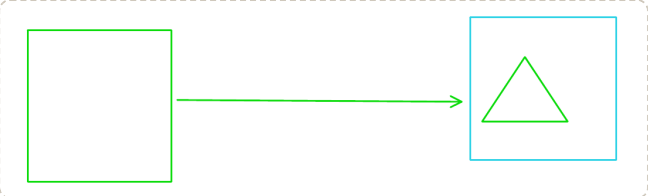






`BankAccount` `EmployeeDetails` `ProductDetails`





Data Type	Default Value (for fields)
`byte`	`0`
`short`	`0`
`int`	`0` ✓
`long`	`0L`
`float`	`0.0f`
`double`	`0.0d`
`char`	`'\u0000'` ✓
`String` (or any object)	`null` ✓
`boolean`	`false`

02 Reference variable:

```
Student s= new Student();
```

> The variable used to refer to an object is called a **reference variable**.

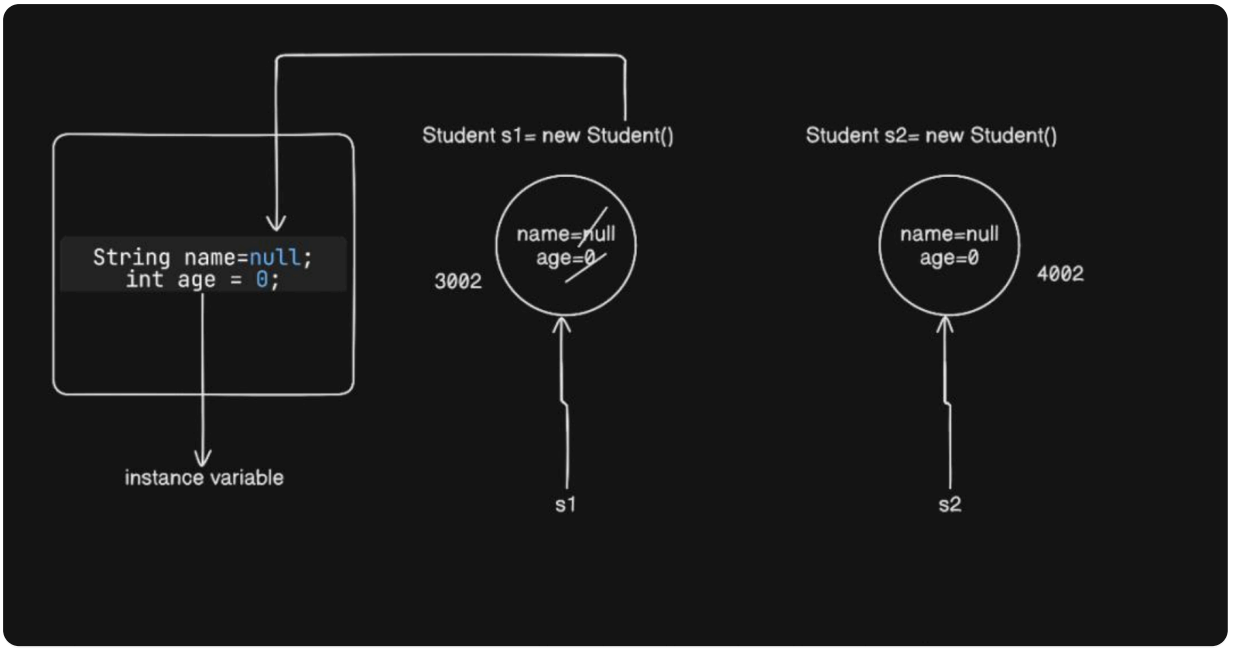
- Here `s` is the reference variable
- `new Student()` creates the object

03 Dot operator

> The **dot operator** is used to access fields and methods through reference

EG:

`s.name`	`s.age`	`s.study()`



04 What is Method ?

> A method is a named block of code that performs a specific task.

Method = reusable block of code that performs an action

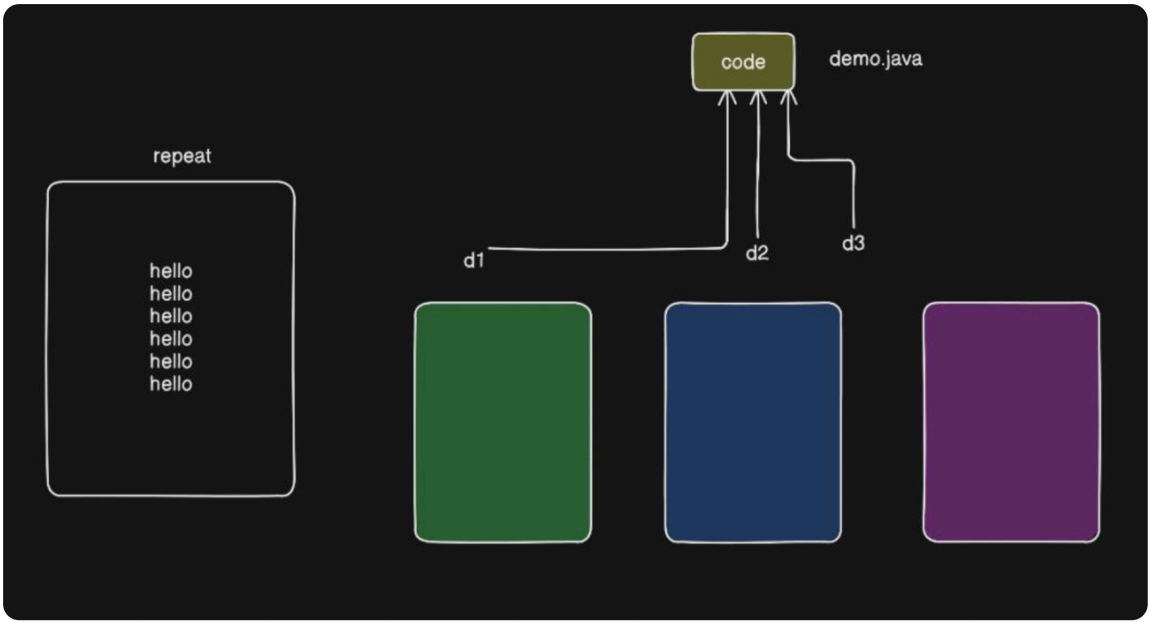
Why we need methods ?

Without method :

```
System.out.println("Name:rahul");
System.out.println("Age:30");
System.out.println("Course: Spark 6.0");
```

If we need this logic in many places then how will we do it ?

We can use method and use it by reference variable



Method declaration

```
void study()
{
    System.out.println("Student is studying.....");
}
```

`void`	`study`	`()`	`{}`
is the return type	method name	contains parameters if any	contains method body

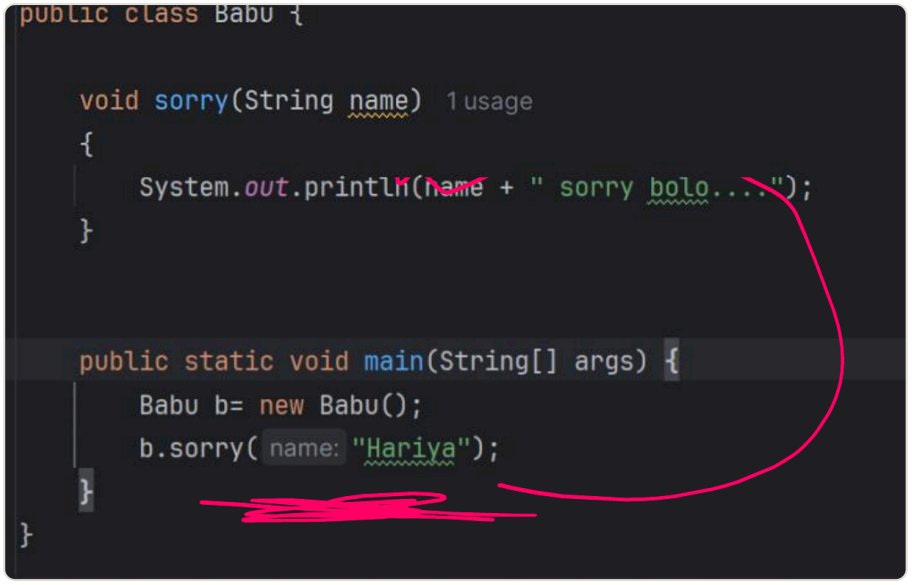
Method call

```
s.study()
```

05 Parameter and Argument

A **parameter** is a variable declared in method definition.

An **argument** is the actual value passed during the method call



06 Method return type

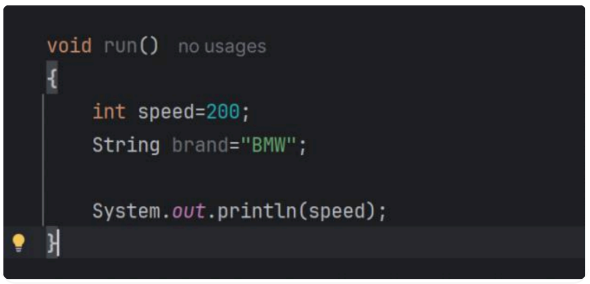
We have learned primitive and non-primitive data types; all of these we can use as method return type.

Four common method types

1 No parameters, No return value	2 Parameter, No return value
3 No parameter, return value	4 Parameter, return value

07 Local variable:

A variable declared inside a method or block is called **local variable**



Note: local variable exists only inside method or block

Instance variable	Local variable
Declared inside class, outside method	Declared inside method or block
Belongs to an object	Belongs to method
Gets default values	Must be initialized before use
Accessible from anywhere in class	Accessible only inside the same method